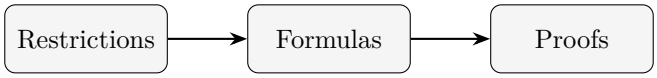
DISCRETE STRUCTURES

CMSC250

Chapter 21: Advanced Counting

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SUBAdvanced Counting



Contents

1 What makes counting advanced? 3

2 Permutations with indistinguishable objects 3

3 Counting multisets: combinations with repetition 3

4 Stars and bars 3

5 The binomial theorem 4

6 Pascal's identity and a combinatorial proof 4

7 Inclusion–exclusion for two and three sets 5

8 The pigeonhole principle 5

9 Derangements: an inclusion–exclusion application 5

10 Practice questions and complete solutions 6

11 Common mistakes 6

1 What makes counting advanced?

Basic counting assumes distinct objects and straightforward choices. More difficult problems have identical objects, repeated selections, constraints, or overlapping cases. The strategy is still to define the outcome carefully, then justify why the formula counts every valid outcome *exactly once*.

2 Permutations with indistinguishable objects

If n objects include n_1 identical copies of one type, n_2 of another, and so on, with $n_1 + \cdots + n_k = n$, then the number of distinct arrangements is

$$\frac{n!}{n_1!n_2!\cdots n_k!}.$$

Why divide? Permuting the n_i identical objects among themselves changes no visible arrangement, yet would be counted $n_i!$ times if all copies were labeled.

Worked example: arrangements of BANANA

Question. How many distinct arrangements are there of the letters in BANANA?

Solution. There are six positions, with three A's, two N's, and one B. Therefore $\frac{6!}{3!2!1!} = 60$.

Worked example: binary strings with a fixed number of ones

Question. How many length-eight bit strings contain exactly three ones?

Solution. Choose the three positions occupied by 1: $\binom{8}{3} = 56$. The remaining five positions must contain 0.

3 Counting multisets: combinations with repetition

Suppose we choose r items from n types, with unlimited copies of each type and order irrelevant. The answer is

$$\binom{n+r-1}{r} = \binom{n+r-1}{n-1}.$$

A selection can be represented by nonnegative counts $x_1 + \cdots + x_n = r$, where x_i is the number of items of type i.

4 Stars and bars

Represent r identical chosen items by stars and separate the n categories using $n-1$ bars. There are $r+n-1$ positions, of which $n-1$ are bars.

Worked example: seven candies in three flavors

Question. How many ways can one choose seven candies from three flavors, with unlimited stock?

Solution. We count nonnegative integer solutions of $x_1+x_2+x_3 = 7$. The number is $\binom{7+3-1}{3-1} = \binom{9}{2} = 36$.

$$\star\,\star \mid \star\,\star\,\star \mid \star\,\star$$

Two of flavor 1, three of flavor 2, two of flavor 3

Worked example: positive integer solutions

Question. How many positive integer solutions satisfy $x_1 + x_2 + x_3 = 10$?

Solution. Since each $x_i \geq 1$, write $y_i = x_i - 1 \geq 0$. Then $y_1 + y_2 + y_3 = 7$. By stars and bars, the answer is $\binom{9}{2} = 36$.

Key idea. Nonnegative solutions to $x_1 + \cdots + x_n = r$: $\binom{r+n-1}{n-1}$. Positive solutions (when $r \geq n$): $\binom{r-1}{n-1}$.

5 The binomial theorem

For any nonnegative integer n,

$$(x + y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$$

The coefficient $\binom{n}{k}$ counts the ways to choose exactly k of the n factors to contribute y, while the other $n - k$ contribute x.

Worked example: find a coefficient

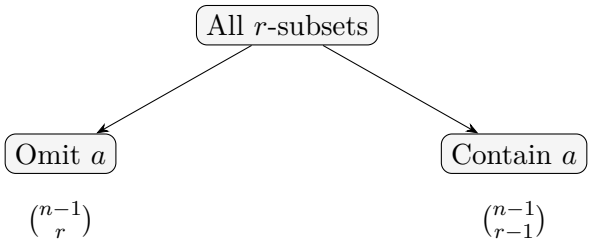
Question. What is the coefficient of x^3y^2 in $(x + y)^5$?

Solution. Choose which two of the five factors contribute y. The coefficient is $\binom{5}{2} = 10$.

6 Pascal's identity and a combinatorial proof

Question. Prove $\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}$ for $1 \leq r \leq n - 1$.

Proof. Fix a particular element a in an n-element set S. Every r-element subset of S either omits a or contains a, and these cases are disjoint. If it omits a, choose all r elements from the remaining $n - 1$, giving $\binom{n-1}{r}$ possibilities. If it contains a, choose the remaining $r - 1$ elements from the other $n - 1$, giving $\binom{n-1}{r-1}$. Adding these disjoint cases proves the identity. ■



7 Inclusion–exclusion for two and three sets

For finite sets A and B,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

For three finite sets,

$$\begin{aligned} |A \cup B \cup C| = {} & |A| + |B| + |C| \\ & - |A \cap B| - |A \cap C| - |B \cap C| \\ & + |A \cap B \cap C|. \end{aligned}$$

A point in all three sets is added three times, subtracted three times, then added once, giving the correct final count of one.

Worked example: students enrolled in three subjects

Question. In a group, 18 study math, 16 study CS, 14 study physics; 7 study both math and CS, 6 both math and physics, 5 both CS and physics; and 3 study all three. How many study at least one?

Solution. Inclusion–exclusion gives $18 + 16 + 14 - 7 - 6 - 5 + 3 = 33$. The pairwise intersections include the three-way intersection, which is why the last term is added back.

8 The pigeonhole principle

If $n + 1$ objects are placed into n boxes, some box contains at least two objects. More generally, if N objects occupy k boxes, at least one box contains at least $\lceil N/k \rceil$ objects.

Worked example: shared birth month

Question. Prove that among 13 people, two were born in the same month.

Proof. There are 12 possible birth months. Assign each of the 13 people to the box representing their birth month. If every month contained at most one person, there could be at most 12 people, contradicting the existence of 13. Therefore some month contains at least two people. ■

9 Derangements: an inclusion–exclusion application

A *derangement* of n labeled objects is a permutation in which no object remains in its original position. Let A_i be the set of permutations fixing object i. By inclusion–exclusion, the number of derangements

is

$$D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

Worked example: derange three letters

Question. How many permutations of ABC have no letter in its original position?

Solution. The valid permutations are BCA and CAB. The formula gives $D_3 = 3!(1-1+1/2-1/6) = 2$.

10 Practice questions and complete solutions

Question. 1. How many distinct arrangements are there of MISSISSIPPI?

Solution. There are 11 letters: four I's, four S's, two P's, and one M. Thus the answer is $\frac{11!}{4!4!2!} = 34650$.
Question. 2. Count the nonnegative solutions of $a + b + c + d = 6$.

Solution. Six stars and three bars give $\binom{9}{3} = 84$ solutions. **Question.** 3. Find the coefficient of x^2y^4 in $(2x + y)^6$.

Solution. Choose two factors to contribute $2x$ and four to contribute y. The coefficient is $\binom{6}{2}2^2 = 60$.
Question. 4. Prove that among 21 integers, two have the same remainder upon division by 20.

Proof. By the quotient–remainder theorem, each integer has exactly one remainder in $\{0, 1, \ldots, 19\}$ when divided by 20. There are 20 possible remainder classes and 21 integers. By the pigeonhole principle, two integers lie in the same class and hence have the same remainder. ■

11 Common mistakes

When items repeat, dividing by factorials is justified only for *indistinguishable* copies. In stars and bars, decide whether zero counts are allowed. In inclusion–exclusion, do not forget the triple-intersection correction. For pigeonhole proofs, identify both the objects and the boxes explicitly.